Mean :

$$E(X) = \sum x f(x) = 40 \cdot \left(\frac{6}{66}\right) + 10 \cdot \left(\frac{32}{66}\right) + (-20) \cdot \left(\frac{28}{66}\right) = 0$$

That is expected winning amount is 0.

Variance :

$$E(X^2) = \sum x^2 f(x) = 40^2 \cdot \left(\frac{6}{66}\right) + 10^2 \cdot \left(\frac{32}{66}\right) + (-20)^2 \cdot \left(\frac{28}{66}\right) = \frac{4000}{11}$$

$$\left(E(X)\right)^2 = 0^2 = 0$$

$$\text{This gives } V(X) = E(X^2) - \left(E(X)\right)^2 = \frac{4000}{11} - 0 = \frac{4000}{11}$$

$$\text{Therefore } E(X) = 0 \text{ and } \text{Var}(X) = \frac{4000}{11}.$$

Example 11.18

Find the mean and variance of a random variable X, whose probability density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Solution

Observe that the given distribution is continuous

Mean :

By definition $\mu = E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$

$$= \int_{-\infty}^{0} 0\left(\lambda e^{-\lambda x}\right) dx + \int_{0}^{\infty} x\left(\lambda e^{-\lambda x}\right) dx$$

(We can also use integration by parts or Bernoulli's formula)

$$= 0 + \lambda \int_{0}^{\infty} x\left(e^{-\lambda x}\right) dx$$

$$= 0 + \lambda \left(\frac{\lfloor 1}{\lambda^2}\right) \left(\text{using Gamma integral for positive integer } n,\ \int_0^\infty x^n e^{-\alpha x} dx = \frac{\lfloor n}{\alpha^{n+1}}\right)$$

$$= \frac{1}{\lambda}$$

Variance :

By definition, $E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\, dx$

$$= \int_{-\infty}^{0} 0\left(\lambda e^{-\lambda x}\right) dx + \int_{0}^{\infty} x^2\left(\lambda e^{-\lambda x}\right) dx$$

(We can also use integration by parts or Bernoulli's formula)

$$= 0 + \lambda \int_{0}^{\infty} x^2\left(e^{-\lambda x}\right) dx$$

$$= 0 + \lambda \left(\frac{\lfloor 2}{\lambda^3}\right) = \frac{2}{\lambda^2} \quad \text{(using Gamma integral for positive integer)}$$

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